
Residual-Based Variational Multiscale Theory of LES Turbulence Modeling*

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Abstract We present an aperçu of our variational multiscale theory of LES turbulence. The theory is succinctly summarized in terms of a finite-dimensional coarse-scale equation governing the resolved scales that depend parametrically upon unresolved fine scales, which in turn are defined in terms of a functional of the coarse-scale residual “lifted” to the dual of the fine-scale space, and the coarse-scale velocity field itself. We illustrate the performance of a numerical implementation of the theory with calculations of a turbulent channel flow at a friction-velocity Reynolds number of 395 and comparisons with the DNS data.

Key words: B-splines, isogeometric analysis, large eddy simulation, turbulence modeling, variational multiscale method, wall-bounded turbulence

1 Variational Multiscale Formulation of the Incompressible Navier–Stokes Equations

1.1 Incompressible Navier–Stokes Equations

We consider a space-time domain $Q = \Omega \times]0, T[\subset \mathbb{R}^3 \times \mathbb{R}^+$ with lateral boundary $P = \Gamma \times]0, T[$, as illustrated in the left-hand side of [Figure 1](#). The initial/boundary-value problem consists of solving the following equations for $\mathbf{u} : \overline{Q} \rightarrow \mathbb{R}^3$, the velocity, and $p : Q \rightarrow \mathbb{R}$, the pressure (divided by the constant density),

*This is an abridged version of Bazilevs et al. [1], which may be consulted for additional implementational details and numerical results.

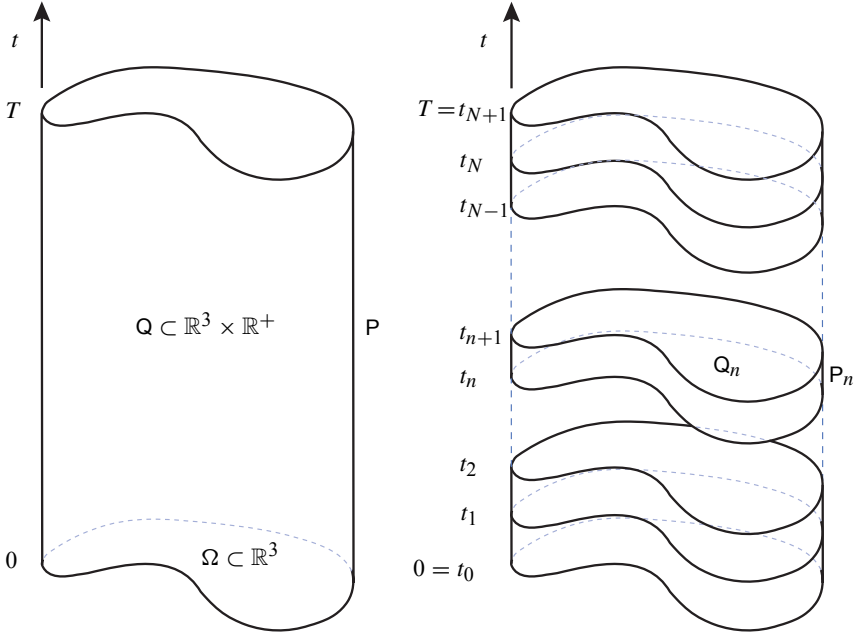


Fig. 1. Space-time domain (left) and slicing into space-time slabs (right).

$$\frac{\partial \mathbf{u}}{\partial t} + \nabla \cdot (\mathbf{u} \otimes \mathbf{u}) + \nabla p = \nu \Delta \mathbf{u} + \mathbf{f} \quad \text{in } Q \quad (1)$$

$$\nabla \cdot \mathbf{u} = 0 \quad \text{in } Q \quad (2)$$

$$\mathbf{u} = \mathbf{0} \quad \text{on } P \quad (3)$$

$$\mathbf{u}(0^+) = \mathbf{u}(0^-) \quad \text{on } \Omega \quad (4)$$

where $\mathbf{f} : Q \rightarrow \mathbb{R}^d$ is the given body force (per unit volume); ν is the kinematic viscosity, assumed positive and constant; $\mathbf{u}(0^-) : \Omega \rightarrow \mathbb{R}^d$ is the given initial velocity; and $\otimes$ denotes the tensor product (e.g., in component notation, $[\mathbf{u} \otimes \mathbf{v}]_{ij} = u_i v_j$).

Equations (1)–(4) are, respectively, the linear momentum balance, the incompressibility constraint, the no-slip boundary condition and the initial condition.

Global Space-Time Variational Formulation

Let $\mathcal{V} = \mathcal{V}(Q)$ denote both the trial solution and weighting function spaces, which are assumed to be identical. We assume $\mathbf{U} = \{\mathbf{u}, p\} \in \mathcal{V}$ implies $\mathbf{u} = \mathbf{0}$ on P and $\int_{\Omega} p(t) d\Omega = 0$ for all $t \in]0, T[$. Let $(\cdot, \cdot)_{\omega}$ denote the L^2 inner product with respect to the domain ω . The variational formulation is stated as follows:

Find $\mathbf{U} \in \mathcal{V}$ such that $\forall \mathbf{W} = \{\mathbf{w}, q\} \in \mathcal{V}$:

$$B(\mathbf{W}, \mathbf{U}) = B_1(\mathbf{W}, \mathbf{U}) + B_2(\mathbf{W}, \mathbf{U}, \mathbf{U}) = L(\mathbf{W}) \quad (5)$$

with

$$B_1(\mathbf{W}, \mathbf{U}) = (\mathbf{w}(T^-), \mathbf{u}(T^-))_{\Omega} - \left(\frac{\partial \mathbf{w}}{\partial t}, \mathbf{u} \right)_{\mathbf{Q}} + (q, \nabla \cdot \mathbf{u})_{\mathbf{Q}} - (\nabla \cdot \mathbf{w}, p)_{\mathbf{Q}} + (\nabla^s \mathbf{w}, 2\nu \nabla^s \mathbf{u})_{\mathbf{Q}} \quad (6)$$

$$B_2(\mathbf{W}, \mathbf{U}, \mathbf{V}) = - (\nabla \mathbf{w}, \mathbf{u} \otimes \mathbf{v})_{\mathbf{Q}} \quad (7)$$

$$L(\mathbf{W}) = (\mathbf{w}, \mathbf{f})_{\mathbf{Q}} + (\mathbf{w}(0^+), \mathbf{u}(0^-))_{\Omega} \quad (8)$$

where $\mathbf{V} = \{\mathbf{v}, \cdot\}$ and $\nabla^s \mathbf{u} = (\nabla \mathbf{u} + (\nabla \mathbf{u})^T) / 2$. Note that $B_1(\cdot, \cdot)$ is a bilinear form and $B_2(\cdot, \cdot, \cdot)$ is a trilinear form. Assuming sufficient regularity and integrating by parts, we obtain the Euler–Lagrange form of (5)–(8):

$$0 = \left(\mathbf{w}, \frac{\partial \mathbf{u}}{\partial t} + \nabla \cdot (\mathbf{u} \otimes \mathbf{u}) + \nabla p - \nabla \cdot 2\nu \nabla^s \mathbf{u} - \mathbf{f} \right)_{\mathbf{Q}} + (q, \nabla \cdot \mathbf{u})_{\mathbf{Q}} + (\mathbf{w}(0^+), \mathbf{u}(0^+) - \mathbf{u}(0^-))_{\Omega} \quad (9)$$

which reveals that the variational formulation implies satisfaction of the momentum equations, incompressibility constraint, and initial condition. The velocity boundary condition is built into the definition of the space $\mathcal{V}$. In summary, the variational formulation is equivalent to (1)–(4).

Sliced Space-Time Variational Formulation

Consider a *slicing* of space-time obtained by replacing $]0, T[$ by $]t_n, t_{n+1}[$, $n = 0, 1, 2, \dots, N$, and summing over the space-time *slabs* $\mathbf{Q}_n$ (see Figure 1). The counterparts of (5)–(9) for a typical slab are:

$$B(\mathbf{W}, \mathbf{U})_n = B_1(\mathbf{W}, \mathbf{U})_n + B_2(\mathbf{W}, \mathbf{U}, \mathbf{U})_n = L(\mathbf{W})_n \quad (10)$$

$$B_1(\mathbf{W}, \mathbf{U})_n = (\mathbf{w}(t_{n+1}^-), \mathbf{u}(t_{n+1}^-))_{\Omega} - \left(\frac{\partial \mathbf{w}}{\partial t}, \mathbf{u} \right)_{\mathbf{Q}_n} + (q, \nabla \cdot \mathbf{u})_{\mathbf{Q}_n} - (\nabla \cdot \mathbf{w}, p)_{\mathbf{Q}_n} + (\nabla^s \mathbf{w}, 2\nu \nabla^s \mathbf{u})_{\mathbf{Q}_n} \quad (11)$$

$$B_2(\mathbf{W}, \mathbf{U}, \mathbf{V})_n = - (\nabla \mathbf{w}, \mathbf{u} \otimes \mathbf{v})_{\mathbf{Q}_n} \quad (12)$$

$$L(\mathbf{W})_n = (\mathbf{w}, \mathbf{f})_{\mathbf{Q}_n} + (\mathbf{w}(t_n^+), \mathbf{u}(t_n^-))_{\Omega} \quad (13)$$

$$0 = \left(\mathbf{w}, \frac{\partial \mathbf{u}}{\partial t} + \nabla \cdot (\mathbf{u} \otimes \mathbf{u}) + \nabla p - \nabla \cdot 2\nu \nabla^s \mathbf{u} - \mathbf{f} \right)_{\mathbf{Q}_n} + (q, \nabla \cdot \mathbf{u})_{\mathbf{Q}_n} + (\mathbf{w}(t_n^+), \mathbf{u}(t_n^+) - \mathbf{u}(t_n^-))_{\Omega} \quad (14)$$

where, in (10)–(14), $\mathbf{U} = \{\mathbf{u}, p\}$ and $\mathbf{W} = \{\mathbf{w}, q\}$ belong to $\mathcal{V}_n = \mathcal{V}(\mathbf{Q}_n)$, the restriction of $\mathcal{V}$ to $\mathbf{Q}_n$. From the Euler–Lagrange form of the equation, (14), we see that the momentum equation and incompressibility constraint are satisfied on

the slab, and the solution is continuous across slab interfaces. The formulation in terms of space-time slabs exploits the causal nature of the Navier–Stokes equations and reduces the overall problem to a succession of initial/boundary-value problems on the slabs. The solution is obtained solving the variational equation on each slab successively, $n = 0, 1, 2, \dots, N$. We emphasize that this is an *exact* formulation, entirely equivalent to (5)–(9), and (1)–(4). However, it is a more suitable starting point for the development of numerical schemes.

Remark

In order to simplify notation, we will work with the global form of the variational equation subsequently. However, all results are equally applicable to the variational equations of the individual space-time slabs.

1.2 Scale Separation

We consider a direct-sum decomposition of $\mathcal{V}$ into “coarse-scale” and “fine-scale” subspaces, $\overline{\mathcal{V}}$ and $\mathcal{V}'$, respectively,

$$\mathcal{V} = \overline{\mathcal{V}} \oplus \mathcal{V}' \quad (15)$$

$\overline{\mathcal{V}}$ is assumed to be a finite-dimensional space and it will be identified later with the space of functions with which we actually compute. In order to make the decomposition well-defined, we need to introduce a procedure for uniquely determining $\overline{\mathbf{U}} \in \overline{\mathcal{V}}$ and $\mathbf{U}' \in \mathcal{V}'$ from a given $\mathbf{U} \in \mathcal{V}$. This can be accomplished with the aid of a projector $\overline{\mathbb{P}} : \mathcal{V} \rightarrow \overline{\mathcal{V}}$. For example, $\overline{\mathbb{P}}$ could be the L^2 -projector, H^1 -projector, etc. There are infinitely many possibilities.¹ Once $\overline{\mathbb{P}}$ is selected, we know how the coarse scales approximate all scales, viz.,

$$\overline{\mathbf{U}} = \overline{\mathbb{P}}\mathbf{U} \quad (16)$$

$$\mathbf{U}' = \mathbf{U} - \overline{\mathbb{P}}\mathbf{U} = (\mathbb{I} - \overline{\mathbb{P}})\mathbf{U} \quad (17)$$

where $\mathbb{I}$ is the identity operator. Likewise, we can decompose a weighting function into its coarse- and fine-scale components:

$$\overline{\mathbf{W}} = \overline{\mathbb{P}}\mathbf{W} \quad (18)$$

$$\mathbf{W}' = \mathbf{W} - \overline{\mathbb{P}}\mathbf{W} = (\mathbb{I} - \overline{\mathbb{P}})\mathbf{W} \quad (19)$$

With these, we may decompose the original variational equation into coupled coarse-scale and fine-scale equations, viz.,

¹The way $\overline{\mathbf{U}}$ is determined from $\mathbf{U}$ is a very important issue, and it has very significant impact on the theory to be developed. An initiatory study of typical projectors is presented in Hughes and Sangalli [8]. Not only can one envision an infinite number of possible projectors, but one can also envision an infinite number of nonlinear optimization schemes that “fit” $\overline{\mathbf{U}}$ to $\mathbf{U}$. In some applications nonlinear schemes will surely be important, an example being compressible turbulence with shocks where monotonicity is important. However, for incompressible turbulence, we feel linear projectors, such as the H^1 -projector, should suffice.

$$B(\overline{\mathbf{W}}, \overline{\mathbf{U}} + \mathbf{U}') = L(\overline{\mathbf{W}}) \quad (20)$$

$$B(\mathbf{W}', \overline{\mathbf{U}} + \mathbf{U}') = L(\mathbf{W}') \quad (21)$$

where

$$\begin{aligned} B(\overline{\mathbf{W}}, \overline{\mathbf{U}} + \mathbf{U}') &= B_1(\overline{\mathbf{W}}, \overline{\mathbf{U}}) + B_1(\overline{\mathbf{W}}, \mathbf{U}') \\ &\quad + B_2(\overline{\mathbf{W}}, \overline{\mathbf{U}}, \overline{\mathbf{U}}) \\ &\quad + B_2(\overline{\mathbf{W}}, \overline{\mathbf{U}}, \mathbf{U}') + B_2(\overline{\mathbf{W}}, \mathbf{U}', \overline{\mathbf{U}}) \\ &\quad + B_2(\overline{\mathbf{W}}, \mathbf{U}', \mathbf{U}') \end{aligned} \quad (22)$$

$$\begin{aligned} B(\mathbf{W}', \overline{\mathbf{U}} + \mathbf{U}') &= B_1(\mathbf{W}', \overline{\mathbf{U}}) + B_1(\mathbf{W}', \mathbf{U}') \\ &\quad + B_2(\mathbf{W}', \overline{\mathbf{U}}, \overline{\mathbf{U}}) \\ &\quad + B_2(\mathbf{W}', \overline{\mathbf{U}}, \mathbf{U}') + B_2(\mathbf{W}', \mathbf{U}', \overline{\mathbf{U}}) \\ &\quad + B_2(\mathbf{W}', \mathbf{U}', \mathbf{U}') \end{aligned} \quad (23)$$

In (22), $B_2(\overline{\mathbf{W}}, \overline{\mathbf{U}}, \mathbf{U}')$ and $B_2(\overline{\mathbf{W}}, \mathbf{U}', \overline{\mathbf{U}})$ correspond to the cross-stress terms, and $B_2(\overline{\mathbf{W}}, \mathbf{U}', \mathbf{U}')$ corresponds to the Reynolds stress term. Equation (21) can be expressed as

$$B_{\overline{\mathbf{U}}}(\mathbf{W}', \mathbf{U}') + B_2(\mathbf{W}', \mathbf{U}', \mathbf{U}') = \langle \mathbf{W}', \mathbf{Res}(\overline{\mathbf{U}}) \rangle_{\mathcal{V}', \mathcal{V}'^*} \quad (24)$$

where

$$\begin{aligned} B_{\overline{\mathbf{U}}}(\mathbf{W}', \mathbf{U}') &= B_1(\mathbf{W}', \mathbf{U}') \\ &\quad + B_2(\mathbf{W}', \mathbf{U}', \overline{\mathbf{U}}) + B_2(\mathbf{W}', \overline{\mathbf{U}}, \mathbf{U}') \end{aligned} \quad (25)$$

$$\langle \mathbf{W}', \mathbf{Res}(\overline{\mathbf{U}}) \rangle_{\mathcal{V}', \mathcal{V}'^*} = L(\mathbf{W}') - B_1(\mathbf{W}', \overline{\mathbf{U}}) - B_2(\mathbf{W}', \overline{\mathbf{U}}, \overline{\mathbf{U}}) \quad (26)$$

in which $\mathbf{Res}(\overline{\mathbf{U}})$ is the coarse-scale residual “lifted” to the dual of the fine-scale space $\mathcal{V}'^*$, $\langle \cdot, \cdot \rangle_{\mathcal{V}', \mathcal{V}'^*}$ is the duality pairing, and

$$B_{\overline{\mathbf{U}}}(\cdot, \mathbf{U}') = \left(\frac{d}{d\varepsilon} B(\cdot, \overline{\mathbf{U}} + \varepsilon \mathbf{U}') \right)_{\varepsilon=0} \quad (27)$$

the linearization of $B(\cdot, \overline{\mathbf{U}} + \mathbf{U}')$ about $\overline{\mathbf{U}}$ in the direction $\mathbf{U}'$. Note that the solution of (24) can be formally represented as a functional of $\overline{\mathbf{U}}$ and $\mathbf{Res}(\overline{\mathbf{U}})$, namely,

$$\boxed{\mathbf{U}' = \mathbf{F}'(\overline{\mathbf{U}}, \mathbf{Res}(\overline{\mathbf{U}}))} \quad (28)$$

The explicit dependence on $\overline{\mathbf{U}}$ in the first argument of $\mathbf{F}'$ emanates from the dependence of the linearized operator $B_{\overline{\mathbf{U}}}$ on $\overline{\mathbf{U}}$. This expression can be inserted into (20) to “close” the finite-dimensional system for $\overline{\mathbf{U}}$,

$$\boxed{B(\overline{\mathbf{W}}, \overline{\mathbf{U}} + \mathbf{F}'(\overline{\mathbf{U}}, \mathbf{Res}(\overline{\mathbf{U}}))) = L(\overline{\mathbf{W}})} \quad (29)$$

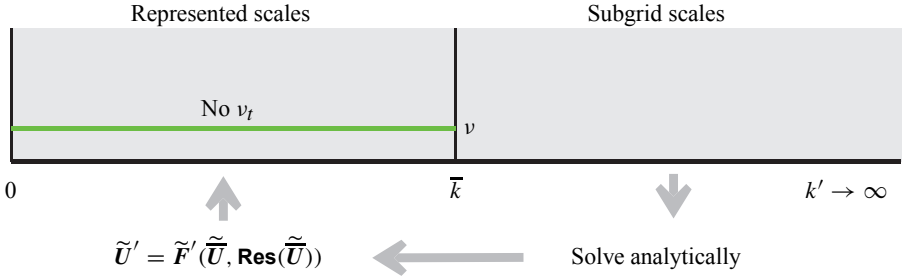


Fig. 2. The variational multiscale turbulence modeling theory is schematically illustrated. The fine, or “subgrid” scales are solved for analytically and substituted into the coarse-scale equation. The coarse scales are the represented scales in a calculation. Note that there is no *ad hoc* eddy viscosity model introduced.

(28) and (29) can be thought of in global terms or in terms of a sequence of space-time slabs. In both cases, they represent a procedure for solving the Navier–Stokes equations in terms of a scale decomposition of the solution. So far we have not discussed approximations or numerics. The solution $\mathbf{U} = \overline{\mathbf{U}} + \mathbf{U}'$, where $\overline{\mathbf{U}}$ is determined by solving (29) and $\mathbf{U}'$ is determined from $\overline{\mathbf{U}}$ through (28), is the *exact* solution of the original variational problem, (20)–(21), and (1)–(4), the Navier–Stokes initial/boundary-value problem.

Our plan for turbulence modeling is to systematically *approximate* the functional $\mathbf{F}'$. This will provide us with a parameterization of the fine scales in terms of the coarse scales, which can be substituted in the coarse-scale equation, “closing” it. The finite-dimensional coarse-scale equation can then be solved. In this way we obtain an approximate coarse-scale solution and an estimation of the fine scales. In summary, our variational multiscale theory of turbulence modeling is encapsulated in the following equations:

$$\boxed{\tilde{\mathbf{U}}' = \tilde{\mathbf{F}}'(\tilde{\mathbf{U}}, \mathbf{Res}(\tilde{\mathbf{U}}))} \quad (30)$$

$$\boxed{B(\overline{\mathbf{W}}, \tilde{\mathbf{U}} + \tilde{\mathbf{F}}'(\tilde{\mathbf{U}}, \mathbf{Res}(\tilde{\mathbf{U}}))) = L(\overline{\mathbf{W}})} \quad (31)$$

where $\tilde{\mathbf{F}}'$ is an approximation of the exact functional $\mathbf{F}'$, and $\tilde{\mathbf{U}}'$ and $\tilde{\mathbf{U}}$ are the approximations of $\mathbf{U}'$ and $\overline{\mathbf{U}}$, respectively. The concept underlying the model is illustrated in Figure 2. We also note that (30) constitutes an *a posteriori* estimation of the *error* in the coarse-scale solution (see [4, 9]).

Remarks

- (31) may be thought of as playing a similar role in the variational multiscale theory as the filtered equations play in traditional turbulence modeling. Distinguishing features are (31) is finite-dimensional and closed, in contrast with the filtered equations.

2. Intuitively, the “better” the fine-scale approximation, the smaller the dimension of the coarse-scale space required, and consequently, the smaller the computational effort. It is also possible to envision a hierarchy of approximations that produce variational multiscale analogues of traditional turbulence modeling concepts, such as large eddy simulation (LES), detached eddy simulation (DES), the Reynolds averaged Navier–Stokes (RANS) approach, etc. LES represents the turbulence modeling methodology requiring the greatest computational burden, but perhaps the least complex modeling. In the following sections we will endeavor to develop a variational multiscale analogue of LES within the theoretical framework of (30) and (31).
3. It is very important to emphasize that in practice we work directly with (31), a finite-dimensional system, and we consider the solution of (31), $\tilde{\bar{U}}$, our approximation to $\bar{U}$, and in turn our approximation to U . Recall, by design of $\mathbb{P}$, $\bar{U}$ is an approximation to U . We do not need to solve for the fine scales and because of this (30) is completely extraneous, unless we wish to use it to estimate the error in the coarse scales. That being said, it may also be interesting to consider $\tilde{\bar{U}} + \tilde{\bar{U}}'$ as an alternative approximation to U . It will of course be necessary to assume that the coarse-scale space is sufficiently large for the philosophy of LES to be appropriate. That is, if there is a well-defined inertial sub-range, then we assume the cut-off between the coarse- and fine-scale spaces resides somewhere within it. This assumption enables us to further assume that the energy content in the fine scales is small compared with the coarse scales, an aspect of considerable importance in attempting to analytically determine the solution of the fine-scale equations.

1.3 Perturbation Series

It seems reasonable to assume that the larger the space $\bar{\mathcal{V}}$, the better the approximation of $\bar{U}$ to U , and the smaller the coarse-scale residual $\mathbf{Res}(\bar{U}) \in \mathcal{V}'^*$. We further assume that if $\mathbf{Res}(\bar{U}) = \mathbf{0}$, then $F'(\bar{U}, \mathbf{0}) = \mathbf{0}$, and if $\mathbf{Res}(\bar{U})$ is “small,” then U' will likewise be “small.”² These assumptions suggest a perturbation series expansion of the form:

$$U' = \varepsilon U'_1 + \varepsilon^2 U'_2 + \varepsilon^3 U'_3 + \cdots = \sum_{k=1}^{\infty} \varepsilon^k U'_k \quad (32)$$

where $\varepsilon = \|\mathbf{Res}(\bar{U})\|_{\mathcal{V}'^*}$. Let us rewrite (24) in terms of the proposed expansion:

$$B_{\bar{U}} \left(W', \sum_{k=1}^{\infty} \varepsilon^k U'_k \right) + B_2 \left(W', \sum_{k=1}^{\infty} \varepsilon^k U'_k, \sum_{k=1}^{\infty} \varepsilon^k U'_k \right) = \varepsilon \langle W', \hat{\mathbf{R}}(\bar{U}) \rangle \quad (33)$$

²These assumptions seem physically reasonable, but rigorous mathematical justification may be difficult to obtain. The existence of nontrivial, unforced weak solutions of the Euler equations, compact in space and time, underscores the mathematical difficulties of the Navier–Stokes equations at large Reynolds numbers (see [11]).

where $\langle \cdot, \cdot \rangle = \langle \cdot, \cdot \rangle_{\mathcal{V}', \mathcal{V}'^*}$, and

$$\widehat{\mathbf{R}}(\overline{\mathbf{U}}) = \frac{\mathbf{Res}(\overline{\mathbf{U}})}{\|\mathbf{Res}(\overline{\mathbf{U}})\|_{\mathcal{V}'^*}} \quad (34)$$

Notice that, by linearity,

$$B_{\overline{\mathbf{U}}} \left(\mathbf{W}', \sum_{k=1}^{\infty} \varepsilon^k \mathbf{U}'_k \right) = \sum_{k=1}^{\infty} \varepsilon^k B_{\overline{\mathbf{U}}} (\mathbf{W}', \mathbf{U}'_k), \quad (35)$$

while the second term requires further consideration. We expand it as follows:

$$\begin{aligned} & B_2 (\mathbf{W}', \varepsilon \mathbf{U}'_1 + \varepsilon^2 \mathbf{U}'_2 + \varepsilon^3 \mathbf{U}'_3 + \cdots, \varepsilon \mathbf{U}'_1 + \varepsilon^2 \mathbf{U}'_2 + \varepsilon^3 \mathbf{U}'_3 + \cdots) \\ &= \varepsilon^2 B_2 (\mathbf{W}', \mathbf{U}'_1, \mathbf{U}'_1) \\ &+ \varepsilon^3 [B_2 (\mathbf{W}', \mathbf{U}'_1, \mathbf{U}'_2) + B_2 (\mathbf{W}', \mathbf{U}'_2, \mathbf{U}'_1)] \\ &+ \varepsilon^4 [B_2 (\mathbf{W}', \mathbf{U}'_1, \mathbf{U}'_3) + B_2 (\mathbf{W}', \mathbf{U}'_2, \mathbf{U}'_2) + B_2 (\mathbf{W}', \mathbf{U}'_3, \mathbf{U}'_1)] \\ &+ \cdots \end{aligned} \quad (36)$$

A recurrence formula can be easily deduced, by grouping coefficients of the powers of ε :

$$\begin{aligned} \varepsilon^2 &\rightarrow B_2 (\mathbf{W}', \mathbf{U}'_1, \mathbf{U}'_1) \\ \varepsilon^3 &\rightarrow B_2 (\mathbf{W}', \mathbf{U}'_1, \mathbf{U}'_2) + B_2 (\mathbf{W}', \mathbf{U}'_2, \mathbf{U}'_1) \\ \varepsilon^4 &\rightarrow B_2 (\mathbf{W}', \mathbf{U}'_1, \mathbf{U}'_3) + B_2 (\mathbf{W}', \mathbf{U}'_2, \mathbf{U}'_2) + B_2 (\mathbf{W}', \mathbf{U}'_3, \mathbf{U}'_1) \\ \varepsilon^5 &\rightarrow \quad \cdots \quad + \quad \cdots \quad + \quad \cdots \quad + \quad \cdots \end{aligned}$$

Hence:

$$B_2 \left(\mathbf{W}', \sum_{k=1}^{\infty} \varepsilon^k \mathbf{U}'_k, \sum_{k=1}^{\infty} \varepsilon^k \mathbf{U}'_k \right) = \sum_{k=2}^{\infty} \varepsilon^k \sum_{j=1}^{k-1} B_2 (\mathbf{W}', \mathbf{U}'_j, \mathbf{U}'_{k-j}) \quad (37)$$

The full expansion of the equation can be compactly written as:

$$\sum_{k=1}^{\infty} \varepsilon^k B_{\overline{\mathbf{U}}} (\mathbf{W}', \mathbf{U}'_k) + \sum_{k=2}^{\infty} \varepsilon^k \sum_{j=1}^{k-1} B_2 (\mathbf{W}', \mathbf{U}'_j, \mathbf{U}'_{k-j}) = \varepsilon \langle \mathbf{W}', \widehat{\mathbf{R}}(\overline{\mathbf{U}}) \rangle \quad (38)$$

Equating like coefficients, we obtain a sequence of linear variational problems coupled through their right-hand sides:

$$\text{For } k = 1 \quad B_{\overline{\mathbf{U}}} (\mathbf{W}', \mathbf{U}'_1) = \langle \mathbf{W}', \widehat{\mathbf{R}}(\overline{\mathbf{U}}) \rangle_{\mathcal{V}', \mathcal{V}'^*} \quad (39)$$

$$\text{For } k \geq 2 \quad B_{\overline{\mathbf{U}}} (\mathbf{W}', \mathbf{U}'_k) = - \sum_{j=1}^{k-1} B_2 (\mathbf{W}', \mathbf{U}'_j, \mathbf{U}'_{k-j}) \quad (40)$$

The bilinear operator $B_{\overline{U}}(\cdot, \cdot)$ is the same for all the problems in the cascade, and can be formally inverted through a Green's operator. The Green's operator concept can be introduced in an abstract sense through a *resolvent* operator:

$$\mathbf{G}'_{\overline{U}}(\cdot) = \mathbf{G}'(\overline{U}, \cdot) : \mathcal{V}'^* \rightarrow \mathcal{V}' \quad (41)$$

$$\mathcal{F}(\cdot) \mapsto \mathbf{V}' \quad (42)$$

such that

$$B_{\overline{U}}(\mathbf{W}', \mathbf{V}') = \mathcal{F}(\mathbf{W}') \quad (43)$$

If a sequence of operators $\mathcal{F}_j : \mathcal{V}' \rightarrow \mathbb{R}$ (i.e., $\mathcal{F}_j \in \mathcal{V}'^*$) is defined as:

$$\begin{aligned} \text{For } k = 1 \quad \mathcal{F}_1(\mathbf{W}') &= \mathcal{F}_1(\mathbf{W}'; \widehat{\mathbf{R}}(\overline{U})) \\ &= \langle \mathbf{W}', \widehat{\mathbf{R}}(\overline{U}) \rangle_{\mathcal{V}', \mathcal{V}'^*} \end{aligned} \quad (44)$$

$$\begin{aligned} \text{For } k \geq 2 \quad \mathcal{F}_k(\mathbf{W}') &= \mathcal{F}_k(\mathbf{W}'; \mathbf{U}'_1, \dots, \mathbf{U}'_{k-1}) \\ &= - \sum_{j=1}^{k-1} B_2 \left(\mathbf{W}', \mathbf{U}'_j, \mathbf{U}'_{k-j} \right) \end{aligned} \quad (45)$$

then it is possible to reformulate the sequence of problems (39)–(40) as:

$$\mathbf{U}'_k = \mathbf{G}'_{\overline{U}}(\mathcal{F}_k) = \mathbf{G}'(\overline{U}, \mathcal{F}_k), \quad k = 1, 2, \dots \quad (46)$$

Notice that in the cascade of problems (39)–(40) (or, equivalently, (44)–(45), or (46)) the level- k term in the expansion depends on terms on the right-hand side, which involve the coarse-scale residual and terms in the expansion from level 1 to $k - 1$.

Upon substituting the $\mathbf{U}'_k$'s into the series (32), the powers of $\varepsilon = \|\mathbf{Res}(\overline{U})\|_{\mathcal{V}'^*}$ cancel out. If the series converges, it represents an exact solution to the fine-scale equation, and then (31) gives the exact solution of the coarse-scale equation. In other words, given the validity of the assumptions, the exact solution of the original Navier–Stokes system is obtained. In order to determine the exact solutions of each of the linear problems in the cascade, we need the exact Green's operator $\mathbf{G}'_{\overline{U}}$. This is a *non-classical* Green's operator, referred to as the “fine-scale Green's operator,” that in turn depends on the *classical* Green's operator and the selected projector $\overline{\mathbb{P}}$ (see Hughes and Sangalli [8]):

$$\mathbf{G}' = \mathbf{G} - \mathbf{G}\overline{\mathbb{P}}^*(\overline{\mathbb{P}}\mathbf{G}\overline{\mathbb{P}}^*)^{-1}\overline{\mathbb{P}}\mathbf{G}, \quad (47)$$

where $\overline{\mathbb{P}}^*$ is the adjoint of $\overline{\mathbb{P}}$. Note that the orthogonality properties

$$\overline{\mathbb{P}}\mathbf{G}' = 0 \quad (48)$$

$$\mathbf{G}'\overline{\mathbb{P}}^* = 0 \quad (49)$$

immediately follow from (47). In Hughes and Sangalli [8] it was shown, in the context of finite element approximations of the advection-diffusion equation, for the

advection-dominated case, that the projector based on the H_0^1 -inner product (termed the Dirichlet projector in Hughes and Oberai [5]) produced a highly *localized* fine-scale Green's operator, despite the classical Green's operator being highly *nonlocal*. In fact, for the one-dimensional case, the support of the fine-scale Green's operator was confined to individual elements, and there was no coupling between elements. It is important to realize that this is not a general feature of the fine-scale Green's operator, but one that depends crucially on the particular projector. For example, the fine-scale Green's operator produced by the L^2 -projector was nonlocal in all cases.

Exact determination of the Green's function is not possible and neither is summing an infinite number of terms in the perturbation series. Consequently, two approximations are necessary in order to develop a practical solution scheme:

1. Approximation of the fine-scale Green's operator for the linearized Navier–Stokes system, $\tilde{\mathbf{G}}'_T \approx \mathbf{G}'_T$.
2. Approximation of the nonlinearities by truncation of the perturbation series.

Once these approximations are made precise, we have defined a turbulence model of the form (31). This will be discussed in the next section.

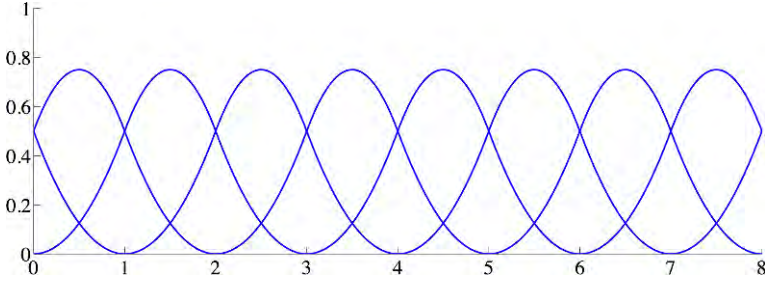
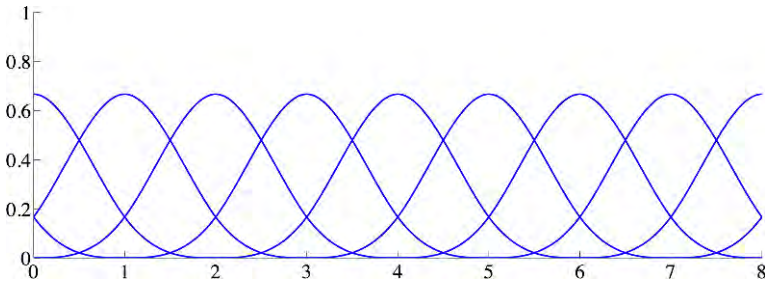
Remark

It needs to be emphasized that the pathway to an approximate turbulence model identified by the above assumptions is not the only possibility, but it does seem a viable candidate for LES-type modeling within the variational multiscale method. Clearly, a more direct attack on the fully nonlinear fine-scale equation, rather than the perturbation series approach, might seem an even more propitious approach. In either case, our theoretical framework for turbulence modeling remains (30) and (31).

2 Turbulent Channel Flow

We consider an equilibrium turbulent channel flow at Reynolds number 395 based on the friction velocity and the channel half width. The computational domain is a rectangular box of size $2\pi \times 2 \times 2/3\pi$ in the stream-wise, wall-normal, and span-wise directions, respectively. A no-slip Dirichlet boundary condition is set at the wall ($y = \pm 1$), while the stream-wise and the span-wise directions are assigned periodic boundary conditions. The no-slip condition is imposed strongly, that is, velocity degrees of freedom are explicitly set to zero at the wall. Alternatively, one may enforce the no-slip conditions weakly by augmenting the discrete formulation with terms that enforce Dirichlet conditions as Euler–Lagrange conditions (see Bazilevs and Hughes [2]). Although the weak boundary condition approach was shown to be superior to the strong imposition, we did not employ it in the computations reported in this paper. The flow is driven by a constant pressure gradient, f_x , acting in the stream-wise direction. The values of the kinematic viscosity ν and the forcing f_x are set to $1.47200 \cdot 10^{-4}$ and $3.372040 \cdot 10^{-3}$, respectively.

The computations were performed on meshes of 32^3 and 64^3 elements. For both meshes we employ C^1 -continuous quadratic, and C^2 -continuous cubic B-Splines.

(a) C^1 -quadratic basis functions.(b) C^2 -cubic basis functions.**Fig. 3.** One-dimensional periodic basis functions.

For both orders, in the stream-wise and the span-wise directions the number of basis functions is equal to the number of elements in these directions (see Figure 3). On the other hand, the number of basis functions in the wall-normal direction is $n_y = n_{el} + p$, where n_{el} is the number of elements in this direction and p is the polynomial order (see Figure 4).

Numerical results for this test case are reported in the form of statistics of the mean stream-wise velocity and root-mean-square velocity fluctuations. Statistics are obtained by sampling the solution fields at the mesh knots and averaging in the stream-wise and span-wise directions as well as in time. Comparison of the statistical quantities of interest with the DNS data of Moser, Kim and Mansour [10] is made in order to assess the accuracy of the proposed turbulence modeling methodology.

Note from Figures 5 and 6 that for a mesh of 64^3 elements both quadratic and cubic solutions are in very close agreement with the DNS results.

The results for 32^3 quadratic and cubic B-splines are even better than high-fidelity spectral Galerkin LES results presented in Hughes et al. [7] and Holmen et al. [3]. We note though that the formulation utilized in [3, 7] employed a fine-scale eddy viscosity model and is quite different from the one used here.

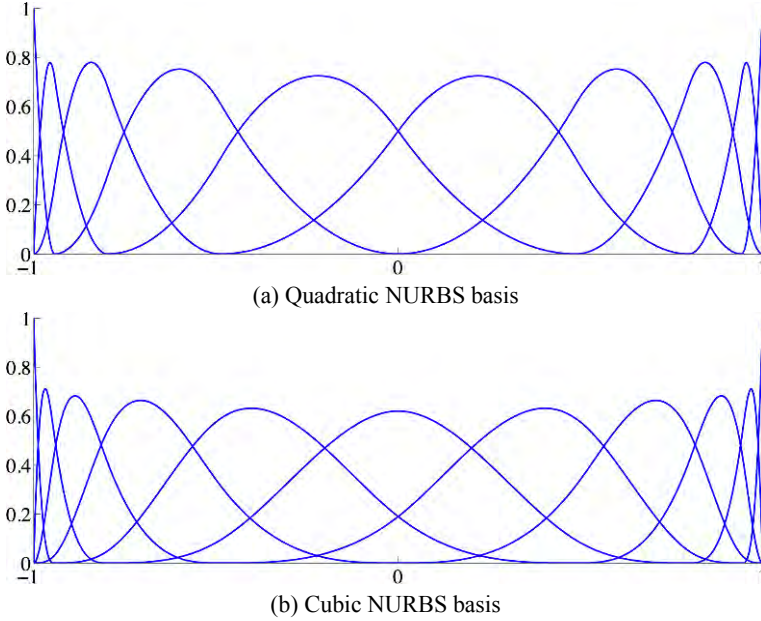


Fig. 4. Illustration of the wall-normal discretization for the turbulent channel flow problem. Meshes are graded towards the ends of the interval in order to better resolve boundary layers. Note that, due to the open knot vector construction (see [6] for details), the first and last basis functions are interpolatory at the endpoints of the domain, which facilitates strong imposition of no-slip Dirichlet boundary conditions.

Figure 7 shows isosurfaces of stream-wise velocity, velocity streamlines, and a series of snapshots of particles released at the channel inflow and set in motion to follow the streamlines in the boundary layer.

3 Conclusions

We presented a general variational multiscale theory suitable for LES-type turbulence modeling. The theory is derived directly from the incompressible Navier–Stokes equations and does not involve any ad hoc mechanisms. In particular, it entirely avoids use of eddy viscosities. We feel that this theory of turbulence modeling is more fundamental and logically consistent than ones derived heretofore and it has significant potential in practical engineering calculations. We might also mention that the turbulence modeling aspects remain *unaltered* when we consider laminar flows. In this sense, our methodology may be viewed as an approach for solving the incompressible Navier–Stokes equations, whether the flow under consideration is laminar or turbulent, or both. We also believe that this aspect separates our theory of turbulence modeling from predecessors.

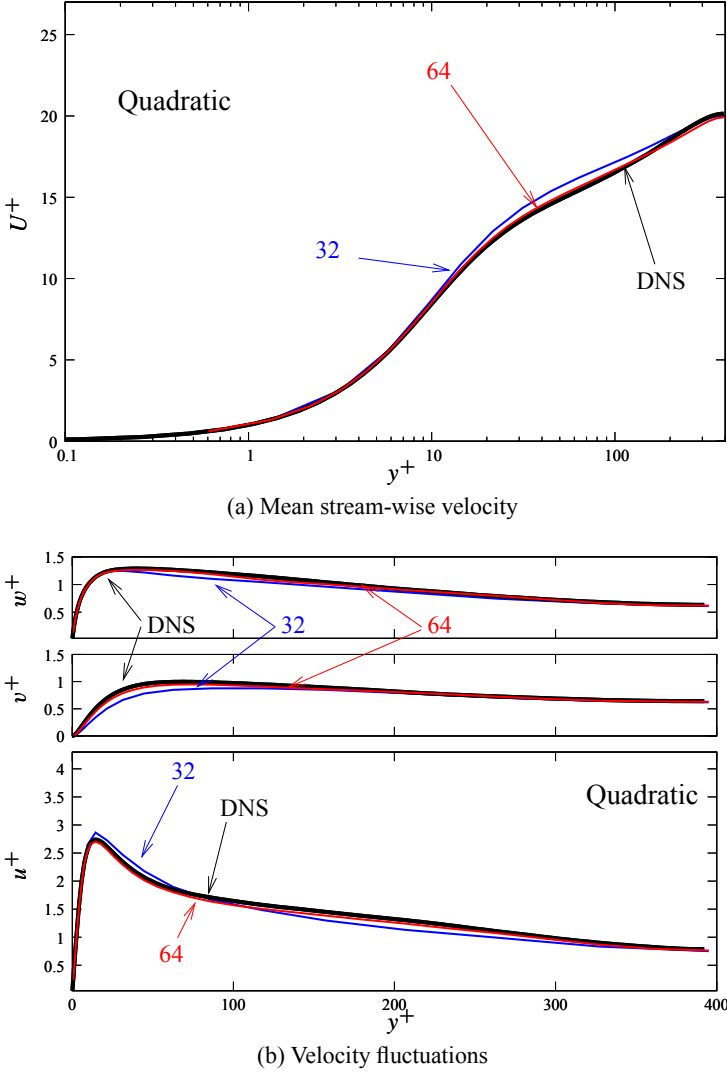


Fig. 5. Turbulent channel flow at $Re_\tau = 395$ computed using quadratic B-splines.

Given that the scale separation in the present methodology is performed with respect to the coarse-scale space actually used in the numerical computations, that is, the resolved scales, and that the fine-scale approximation is rendered well-defined by a projector used to make precise the direct sum decomposition into coarse and fine scales, it is impossible to entirely separate modeling and numerical concepts. We accept this as a fact associated with correct LES-type modeling concepts, not a shortcoming.

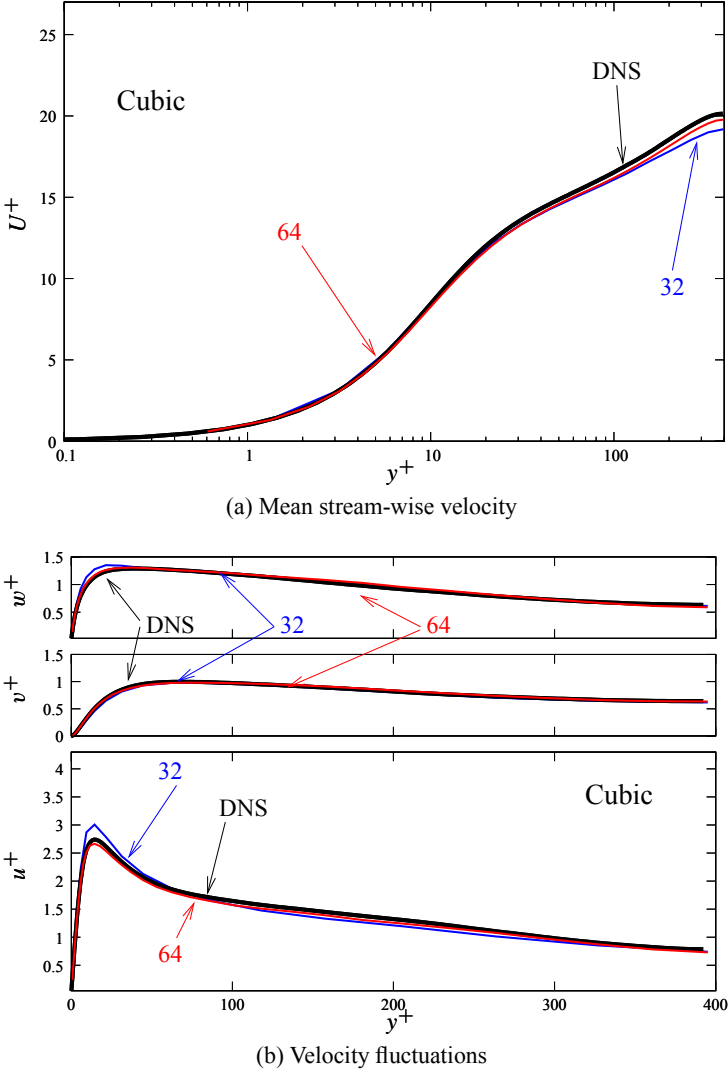


Fig. 6. Turbulent channel flow at $Re_\tau = 395$ computed using cubic B-splines.

We also believe that our theory is more coherent mathematically than previous formulations and that it may be possible to use it as a basis of a statistical analysis of convergence and approximation. This would represent a very significant step forward for the theory of turbulence modeling, but, admittedly, a very difficult one to achieve. Nevertheless, we feel a door has been opened for the construction of a mathematical theory.

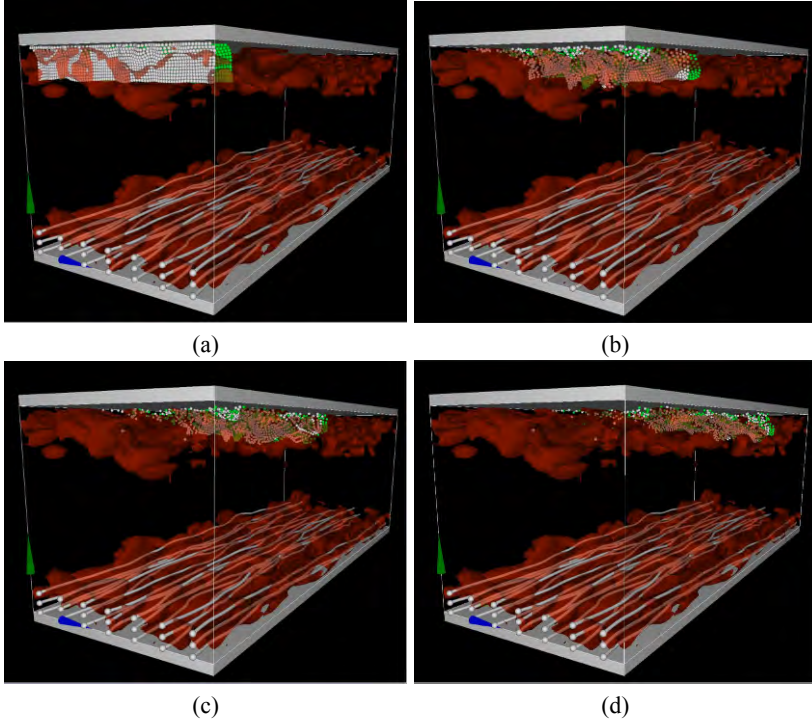


Fig. 7. Turbulent channel flow at $Re_\tau = 395$. Flow streamlines and isosurfaces of streamwise velocity. Two parallel planes of particles are released at the channel inflow close to the wall (see (a)), and are set in motion to follow the streamlines. Snapshots of the particle field are shown as particles travel down the length of the channel (see (b)-(d)). One can see the formation of the boundary layer as the particles released closer to the wall travel at slower speeds compared to the ones released in the outer layer. Also note that the particles initially released on different planes are mixed together as they approach the outflow, revealing the presence of faster and slower streaks in the boundary layer. Solution on the mesh of 32^3 quadratic NURBS was used for this visualization.

In summary, we feel a new paradigm for turbulence modeling has been established. Initial results seem to indicate its accuracy per degree of freedom is superior or, at the least, equal to any procedure proposed heretofore. Its generality and geometric flexibility also suggest it may provide a more powerful approach to turbulence calculations than previously existed.

Acknowledgements

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